

■ Mathematics — Free Learning Resources

YouTube Channels • Websites • Practice Questions — Bilkul Muft!

Ye PDF Mathematics Syllabus ki companion guide hai. Isme har level ke liye **free online resources** diye gaye hain — YouTube videos, websites, aur practice question platforms. Ye sab resources ya to **Hindi/Urdu** mein hain ya English mein (jo internationally sabse best hain). Koi bhi course kharidne ki zaroorat nahi — ye sab **100% muft** hain.

LEVEL	BEST YOUTUBE CHANNEL	BEST WEBSITE
3–8 Saal	Khan Academy Kids	khanacademy.org
8–14 Saal	Khan Academy (Hindi)	khanacademy.org/math
14–17 Saal	Vedantu / Unacademy	brilliant.org (free tier)
BSc Level	MIT OpenCourseWare	ocw.mit.edu
MSc/PhD	NPTEL / Harvard Lectures	math.harvard.edu

SECTION 1: 3–8 Saal — Choti Umar Ke Bacche

■ YouTube Channels (Hindi/Urdu mein)

■ Khan Academy Kids (Urdu/Hindi)

■ youtube.com/@KhanAcademyUrdu

Ginti, shapes, patterns — bilkul asaan, animated. 3-8 saal ke liye perfect.

■ Letstute Elementary Math

■ youtube.com/@Letstute

Class 1-5 tak ke topics: Ginti, jorna, ghatana — Hinglish mein.

■ Maths Ki Duniya

■ youtube.com/@mathskiduniya

Chhoti kaksha ke liye basic maths — Hindi mein seedha aur saaf.

■ CoolMath4Kids

■ youtube.com/@CoolMathGames

Games aur puzzles ke zariye maths sikhna — bacche ko pasand aayega.

■ NumberBlocks (BBC)

■ youtube.com/@Numberblocks

3-6 saal ke liye: numbers ko cartoon characters ki tarah dikhata hai. Bohot effective!

■ Websites aur Apps

■ Khan Academy (Hindi) — khanacademy.org/math

Har topic ka video + practice question. Hindi mein available. Account banakar progress track karo.

■ Splash Learn — splashlearn.com

Grade 1-5 tak ke liye games wali maths. Bacche khelte khelte sikhte hain.

■ Math Playground — mathplayground.com

Free math games for ages 6-12. No signup needed.

■ IXL Math — ixl.com/math

Grade level ke mutabiq practice questions. Free tier mein 10 questions daily.

■ Toppr (Hindi) — toppr.com

Class 1 se 12 tak NCERT-based content. Hindi mein bhi available.

■ Practice Questions ke Resources

■ NCERT Books Free — ncert.nic.in

Class 1 se 12 tak ki NCERT books bilkul free download karo. Best Indian curriculum.

■ Math Worksheets 4 Kids — mathworksheets4kids.com

Thousands of free printable worksheets. Age/grade filter lagao.

■ K5 Learning — k5learning.com

Grade K-5 ke liye free worksheets. PDF mein download karo aur print karo.

■ *Parents ke liye tip: Bacche ko roz sirf 15-20 minute practice karwao. Jab sahi jawab de toh shabaash zaroor kaho!*

SECTION 2: 8–14 Saal — Middle School Level

■ YouTube Channels

- Khan Academy Math (Hindi) — youtube.com/@khanacademyhindi
Class 1-12 tak ke sabhi topics. Fractions, Algebra, Geometry — sab kuch Hindi mein. SABSE BEST resource.
- Vedantu Math (Hindi) — youtube.com/@VedantuMath
NCERT Class 6-10 ke saare topics cover karta hai. Live classes bhi hain.
- BYJU'S — youtube.com/@byjus
Animated videos se concepts clear hote hain. Class 4-12 available.
- Maths Wallah (PW) — youtube.com/@MathsWallah
Alakh Pandey Sir ki maths — bohot asaan aur interesting. Class 9-12 focus.
- Eddie Woo (English) — youtube.com/@misterwootube
Australian teacher — best geometry aur calculus explanations. Class 8+ ke liye.
- PatrickJMT (English) — youtube.com/@patrickjmt
Step-by-step solutions. Algebra se Calculus tak. Bahut saare topics.
- Professor Leonard (English) — youtube.com/@ProfessorLeonard
Full lecture series: Pre-Algebra se Calculus tak. Ek class ki tarah.

■ Websites

- Khan Academy — khanacademy.org/math
Complete curriculum Classes 6-10. Mastery-based learning. Bilkul free, account zaroori.
- NCERT Solutions — ncert.nic.in / ncertsolutions.com
Har NCERT problem ka step-by-step solution. Board exams ke liye must.
- Mathway — mathway.com
Koi bhi math problem type karo — step-by-step solution milega. Algebra, Geometry, etc.
- GeoGebra — geogebra.org
Interactive geometry tool. Shapes banao, graphs plot karo. Bilkul free.
- Desmos — desmos.com
Best free graphing calculator. Equations ka graph dekhna. Class 8 se upar.
- Brilliant.org (Free tier) — brilliant.org
Problem solving aur puzzles. Free tier mein daily problems milte hain.
- Art of Problem Solving — artofproblemsolving.com
Math olympiad ki tayyari. Challenging problems aur detailed solutions.

■ Practice Questions

- NCERT Books (Class 6-10) — ncert.nic.in
India mein Class 6-10 ke liye official, free textbooks. Exercises solve karna best practice hai.
- RD Sharma Solutions — rdsharmamathsbook.com
NCERT se zyada challenging problems. Free solutions available.
- Vedic Maths Problems — vedicmaths.org
Mental math tricks aur fast calculation ke liye.

■ Olympiad Problems — imo-official.org

International Math Olympiad ke past papers. Challenging!

■ Toppr Free Questions — toppr.com

NCERT + extra questions. Class 5-12 covered. Hindi medium bhi.

■ *Tip: Roz ek 'challenge problem' zaroor karo. Sirf book se nahi, AoPS ya Olympiad problems bhi try karo!*

SECTION 3: 14–17 Saal — High School / Class 10-12

■ YouTube Channels

- Maths Wallah Alakh Pandey — youtube.com/@MathsWallah
Class 11-12 Physics Wallah founder ki maths. JEE level tak. Bahut popular.
- Neha Agrawal Mathematically Inclined — youtube.com/@mathematicallyinclined
Class 12 + JEE + Board prep. Calculus, Algebra sab covered.
- 3Blue1Brown — youtube.com/@3blue1brown
Visual mathematics — Calculus, Linear Algebra, Number Theory. English mein. Mind-blowing!
- The Organic Chemistry Tutor (Math) — youtube.com/@TheOrganicChemistryTutor
Pre-Calculus, Calculus, Algebra II. Very detailed. English.
- Professor Leonard — youtube.com/@ProfessorLeonard
Full Calculus I, II, III courses — University level ke jaisa. Best free course.
- Blackpenredpen — youtube.com/@blackpenredpen
Advanced Calculus problems, integration tricks. Class 12 + aage ke liye.
- BYJU'S JEE — youtube.com/@ByjusJEE
JEE Advanced level maths. Advanced topics ke liye.

■ Websites

- Khan Academy Calculus — khanacademy.org/math/calculus-1
Calculus AP level. Videos + practice. Ek dum seedha explanation.
- Paul's Online Math Notes — tutorial.math.lamar.edu
Best text-based notes for Calculus I, II, III, DE. Free aur detailed.
- Wolfram Alpha — wolframalpha.com
Koi bhi equation solve karo. Steps bhi dikhata hai. Calculus ke liye amazing.
- Symbolab — symbolab.com
Step-by-step math solver — derivatives, integrals, equations. Free.
- MIT OpenCourseWare (preview) — ocw.mit.edu/courses/18-01sc
Single Variable Calculus — MIT ka free course. Lecture notes + problems.
- Brilliant.org — brilliant.org
Interactive math problems. Calculus, Pre-calc, Algebra.
- Art of Problem Solving — artofproblemsolving.com
Pre-Olympiad aur competition math. Counting, Number Theory, Geometry advanced.

■ Practice Questions

- JEE Past Papers — jeemain.nta.nic.in
India ki toughest exam ke past papers free mein. Class 12 level se upar.
- CBSE Sample Papers — cbse.gov.in
Official CBSE ke past papers aur sample papers. Free download.
- Cambridge A-Level Past Papers — cambridgeinternational.org
International level — A-Level Math past papers. World standard.

■ MIT OCW Problem Sets — ocw.mit.edu

MIT ke actual homework aur exam problems. Free.

■ AoPS Alcumus — artofproblemsolving.com/alcumus

Adaptive problem system. Har problem solve karne par agle level ka milta hai.

■ *Tip: 3Blue1Brown ke 'Essence of Calculus' aur 'Essence of Linear Algebra' playlists zaroor dekho — concepts kabhi nahi bhoolenge!*

SECTION 4: 18–21 Saal — Undergraduate / BSc Level

■ YouTube Channels

- MIT OpenCourseWare — youtube.com/@mitocw
Gilbert Strang ki Linear Algebra, Strang ki Calculus. MIT ke actual professors ki lectures.
- 3Blue1Brown — youtube.com/@3blue1brown
Linear Algebra, Calculus visual series. Abstract concepts ko picture se samjhao.
- Professor Leonard — youtube.com/@ProfessorLeonard
Calculus II, III, Differential Equations full courses.
- Mathologer — youtube.com/@Mathologer
Deep mathematical ideas ko entertaining way se. Number Theory, Geometry.
- NPTEL Mathematics — youtube.com/@nptel
IIT professors ki lectures. Real Analysis, Abstract Algebra, Topology. Hindi bhi available.
- Harvard Mathematics — youtube.com/@HarvardMathematics
Harvard ke public math lectures.
- Michael Penn — youtube.com/@MichaelPennMath
Abstract Algebra, Real Analysis, Number Theory. University level. Very clear.
- Insights into Mathematics — youtube.com/@njwildberger
Foundations of mathematics, History of math. Alternative perspectives.

■ Websites aur Online Courses

- MIT OpenCourseWare — ocw.mit.edu/mathematics
MIT ke sabhi math courses FREE. Lecture notes, problem sets, exams — sab kuch. World's best resource.
- Paul's Online Math Notes — tutorial.math.lamar.edu
Calculus I/II/III, Linear Algebra, Differential Equations. Best text notes.
- Abstract Algebra (Dana Ernst) — danaernst.com
Free Abstract Algebra notes. Groups, Rings, Fields clearly explained.
- Real Analysis (free book) — classicalrealanalysis.info
Free textbook for Real Analysis. With exercises.
- Linear Algebra Done Right — linear.axler.net
Axler ki famous book ki first edition online free padhi ja sakti hai.
- Coursera (Audit for free) — coursera.org
Audit karo aur lectures dekho bilkul free. Johns Hopkins, Yale, Stanford ki courses.
- edX (Audit free) — edx.org
MIT, Harvard ke courses audit karo. Free mein videos dekho.
- Stacks Project — stacks.math.columbia.edu
Algebraic Geometry ka complete reference. Advanced.
- nLab — ncatlab.org
Category Theory, Topology, Mathematical Physics. Advanced reference.
- Proof Wiki — proofwiki.org
Mathematica theorems ke proofs. Wikipedia jaisi but sirf proofs.

■ Free Textbooks

■ Library Genesis — libgen.is

Duniya ki sari math books FREE mein milti hain. Rudin, Dummit & Foote, Munkres — sab.

■ Open Textbook Library — open.umn.edu/opentextbooks

Peer-reviewed free textbooks. Calculus, Linear Algebra, Statistics.

■ Project Gutenberg Math — gutenberg.org

Classic math books. Old but gold.

■ Springer Free Books — link.springer.com

COVID ke baad kai Springer books free ho gayi hain. GTM series included.

■ Practice Resources

■ MIT OCW Problem Sets — ocw.mit.edu

MIT ke actual homework problems. Bahut challenging. Solutions bhi available.

■ Paul's Notes Practice — tutorial.math.lamar.edu

Har chapter ke baad practice problems + solutions.

■ Math Stack Exchange — math.stackexchange.com

Koi bhi doubt poocho — community jawab deti hai. Lakhon solved problems.

■ NPTEL Assignment Problems — nptel.ac.in

IIT courses ke assignments free mein.

■ *Tip: MIT OCW ke 18.01, 18.02, 18.06, 18.100 courses se shuroo karo — ye world standard hain aur bilkul free hain!*

SECTION 5: MSc / PhD Level — Research Mathematics

■ YouTube Channels (Advanced)

- IHES — youtube.com/@IHES_FR
Institut des Hautes Etudes Scientifiques. World's top mathematicians ki talks.
- ICTP Mathematics — youtube.com/@ICTPMathematics
International Centre ke graduate level lectures. Free, public.
- MSRI / SLMATH — youtube.com/@SLMath_Berkeley
Berkeley ka math research institute. Workshops aur seminars online.
- Talks @ Google Math — youtube.com/@TalksAtGoogle
Top mathematicians Google pe aakar talks dete hain.
- Aleph 0 — youtube.com/@Aleph0
Graduate-level topics: Topology, Algebraic Geometry — beautifully explained.
- Richard Borcherds — youtube.com/@RichardEBorcherds
Fields Medal winner ki lectures — Algebra, Algebraic Geometry, Number Theory.
- Oxford Mathematics — youtube.com/@OxfordMathematics
Oxford ke public lectures. Graduate aur research level.

■ Research Resources

- arXiv Mathematics — arxiv.org/archive/math
Duniya ki sabhi latest math research papers FREE. Research track karne ke liye must.
- MathSciNet — mathscinet.ams.org
Math papers ka database. University se free access milta hai.
- zbMATH Open — zbmath.org
Free math research database.
- Perimeter Institute — perimeterinstitute.ca/videos
Mathematical Physics ki best lectures. Free recordings.
- Harvard Math Department — math.harvard.edu
Seminar schedules, course materials. Public access.
- PROMYS — promys.org/promys-europe
Advanced summer math program for talented students. Partial scholarships.
- Codeforces / Project Euler — projecteuler.net
Mathematical programming problems. Algorithms aur math ka sangam.

■ Recommended Graduate Books (Free Sources se)

- Baby Rudin (Principles of Mathematical Analysis) — libgen.is
Real Analysis ki bible. MSc level. Har mathematician ne padhi hai.
- Dummit & Foote (Abstract Algebra) — libgen.is
Abstract Algebra ka sabse comprehensive book. Groups, Rings, Fields, Modules.
- Munkres (Topology) — libgen.is
Topology ka standard textbook. Point-set se Algebraic Topology.

■ Hartshorne (Algebraic Geometry) — libgen.is

Algebraic Geometry ka classic reference. PhD level.

■ Lang (Algebra) — libgen.is

Graduate Algebra. Serge Lang ka encyclopedic reference.

■ Evans (PDEs) — libgen.is

Graduate PDEs ka standard text.

■ Bredon (Topology and Geometry) — libgen.is

Combined Algebraic Topology aur Differential Geometry.

SECTION 6: Parents Ke Liye — Ghar Par Kaise Padhao?

■ Daily Routine Banao

Roz ek fixed time dedo maths ke liye. 3-8 saal: 15-20 min. 8-14 saal: 30-45 min. 14+ saal: 1-2 ghante. Consistency se zyada koi cheez kaam nahi karti.

■ Sahi Apps Use Karo

Khan Academy Kids (under 8), Khan Academy (8+), Photomath (step-by-step solver), GeoGebra (graphs), Desmos (calculator) — ye sab muft hain.

■ Concept Clear Karo, Marks Nahi

Pehle WHY samajhna zaroori hai. Agar $5 \times 3 = 15$ hai, toh bacche ko samajhna chahiye ke 5 ko 3 baar jodte hain. Rote learning se door raho (tables ke alawa).

■ Olympiad Mein Hissa Lo

IMO (International), AMC (American), NMTC (India), SOF Olympiad mein participate karo. Ye sab skills sharpen karta hai aur confidence badhata hai.

■ Sawaal Poochna Sikhao

Bacche ko kaho ke 'kyun' pooche. ' $x+5=12$ mein $x=7$ kyun?' Is sawaal ka jawab dene se concept bohot mazboot hota hai.

■ Mushkil Problems Se Mat Bhago

Ek problem 20-30 minute mein nahi hoti toh normal hai. Sirf answers dekhne se nahi seekhte — khud try karna zaroori hai.

■ Galtiyan Celebrate Karo

Galti karna seekhne ka hissa hai. Galat jawab dekh ke daanto nahi — 'Chalo milke sochen kahan galti hui' — ye approach zyada effective hai.

■ Milestones Track Karo

Pehli PDF ke syllabus ko dekho aur har mahine check karo ke baccha kis stage par hai. Upar ya neeche — koi judgment nahi, sirf awareness.

Quick Reference: Top 10 Resources

#	Resource	Level	Link
1	Khan Academy Hindi	3–17 saal	khanacademy.org
2	NCERT Books	6–17 saal	ncert.nic.in
3	MIT OpenCourseWare	BSc+	ocw.mit.edu
4	3Blue1Brown	14 saal+	youtube.com/@3blue1brown
5	Professor Leonard	14 saal+	youtube.com/@ProfessorLeonard
6	NPTEL Math	BSc+	nptel.ac.in
7	Paul's Math Notes	14 saal+	tutorial.math.lamar.edu
8	arXiv Math	MSc/PhD	arxiv.org/math
9	Art of Problem Solving	12 saal+	artofproblemsolving.com
10	GeoGebra	8 saal+	geogebra.org

Ye do PDFs mila ke ek mukammal self-learning system hai. Pehli PDF mein KYA padhna hai, aur ye PDF mein KAHAN se padhna hai. Shubh Aaghaz! Mathematics ka safar mushkil zaroor hai, lekin bohot khubsoorat bhi hai.